SIMATS ENGINEERING

ASSESSMENT TOOL 3: Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I A.Shanbagam /192126016o (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A.Shanbagam-A

Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

Here are 10 additional technical presentation topics in a similar style, suitable for a 6–8 minute presentation with definitions, architecture, industry relevance, and a practical example:

1. **Containerization and Cloud-Native Applications: Transforming Modern Software Deployment**
   - Explore containers, container architecture, benefits, and a real-world deployment example.
2. **Serverless Computing: Reducing Infrastructure Management in the Cloud**
   - Discuss Function-as-a-Service (FaaS), architecture, business advantages, and practical use cases.
3. **Edge Computing vs Cloud Computing: Choosing the Right Computing Paradigm**
   - Compare architectures, applications, and industry adoption scenarios.
4. **Microservices Architecture in Cloud Environments**
   - Explain microservices design, benefits over monolithic systems, and industry implementations.
5. **Cloud Security Fundamentals: Protecting Data and Applications**
   - Cover security models, shared responsibility architecture, and real-world security practices.
6. **Multi-Cloud and Hybrid Cloud Strategies for Enterprise IT**
   - Analyze deployment models, business benefits, challenges, and practical examples.

7. **The Role of Artificial Intelligence and Machine Learning in Cloud Platforms**
    - Discuss AI/ML cloud services, architecture, and industry applications.
8. **DevOps and Continuous Integration/Continuous Deployment (CI/CD) in Cloud Computing**
    - Explain DevOps principles, CI/CD architecture, and practical implementation examples.
9. **Cloud Storage Technologies: Object, Block, and File Storage Compared**
    - Compare storage architectures, use cases, and industry adoption patterns.
10. **Internet of Things (IoT) and Cloud Computing: Building Connected Ecosystems**
    - Examine IoT-cloud architecture, benefits, challenges, and real-world applications.
11. **Cloud Migration Strategies: Moving Enterprise Applications to the Cloud**
    - Examine migration approaches, cloud architecture considerations, benefits, challenges, and a successful migration case study.
12. **Big Data Analytics in Cloud Computing**
    - Explore cloud-based data processing architectures, analytics platforms, business value, and real-world applications.
13. **Identity and Access Management (IAM) in Cloud Environments**
    - Discuss authentication, authorization architectures, security benefits, and practical enterprise implementations.
14. **Content Delivery Networks (CDNs) and Their Role in Cloud Services**
    - Explain CDN architecture, performance optimization, industry relevance, and examples from popular web platforms.
15. **Server Virtualization and Its Role in Cloud Computing**

    Explain virtualization architecture, hypervisors, virtual machines, resource allocation, and the benefits of virtualization in cloud data centers.

16. **Cloud Storage Technologies and Architecture**

    Discuss cloud storage concepts, storage architectures, scalability, availability, data replication, and practical applications such as Google Drive and Amazon S3.

17. **Cloud Security: Threats and Protection Mechanisms**

    Explain common cloud security threats, data protection, encryption, network security, security controls, and practical approaches used by cloud service providers.

18. **Infrastructure as a Service (IaaS): Architecture and Applications**

    Explain the IaaS model, virtual computing resources, storage and networking services, advantages, limitations, and practical enterprise applications.

19. **Platform as a Service (PaaS) for Cloud Application Development**

    Discuss PaaS architecture, development environments, application deployment, scalability, advantages, limitations, and examples of PaaS platforms.

20. **Software as a Service (SaaS): Architecture and Business Applications**

    Explain SaaS architecture, features, advantages, subscription-based software delivery, security considerations, and examples such as Google Workspace and Microsoft 365.

Minimum slide flow:

- Title
- Aim and objective
- Core technical explanation
- Architecture or process diagram
- Methodology
- Real-world use case
- Conclusion and key takeaway

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism

Topic :

Cloud Storage Technologies and Architecture:

- Cloud storage stores data on remote servers accessed through a network.
- Users and applications access stored data through interfaces such as APIs.
- Main storage models: object storage, block storage and file storage.
- Important design goals: scalability, availability, durability and security.
- This presentation focuses on the storage workflow, replication and Amazon S3.

Objective

- Differentiate object, block and file storage.
- Describe scalability, availability and data durability.
- Trace the upload, storage, replication and retrieval process.
- Identify security controls and practical applications.

Core technical explanation:

Problem statement:

- Applications need reliable storage that can grow as data volume increases.
- Data must remain accessible and protected against loss and unauthorized access.

Cloud storage concept:

- Data is stored on remote infrastructure managed through network-based access.
- APIs connect users/applications to storage services.
- Metadata helps describe and locate stored data; access controls regulate permissions.
- Distributed storage nodes and replication support availability and durability.

Main storage models:

- Object storage: stores data as objects with identifiers and metadata; suitable for files, media and backups.
- Block storage: stores data in fixed-size blocks; commonly used for virtual-machine disks and databases.
- File storage: organizes data in files and folders; useful when applications need a shared file system.

Architecture or process diagram:

User / Application
↓ Upload or retrieval request
Network / API endpoint
↓
Authentication and access control
↓
Metadata and request handling
↓
Distributed storage nodes ⇄ Replicated copies
↓
Stored data returned to the authorized user/application

Note: This is a simplified conceptual flow; the exact internal design depends on the cloud provider

Real-world use case:

Amazon Simple Storage Service (Amazon S3)

- Amazon S3 is a cloud object-storage service.
- Applications can upload data and retrieve it when required.
- Typical uses include storing images, videos, documents, backups and application data.
- The cloud-storage design supports growth in data volume and access through service interfaces.
- Access permissions and other security controls should be configured to protect stored data

Google Drive

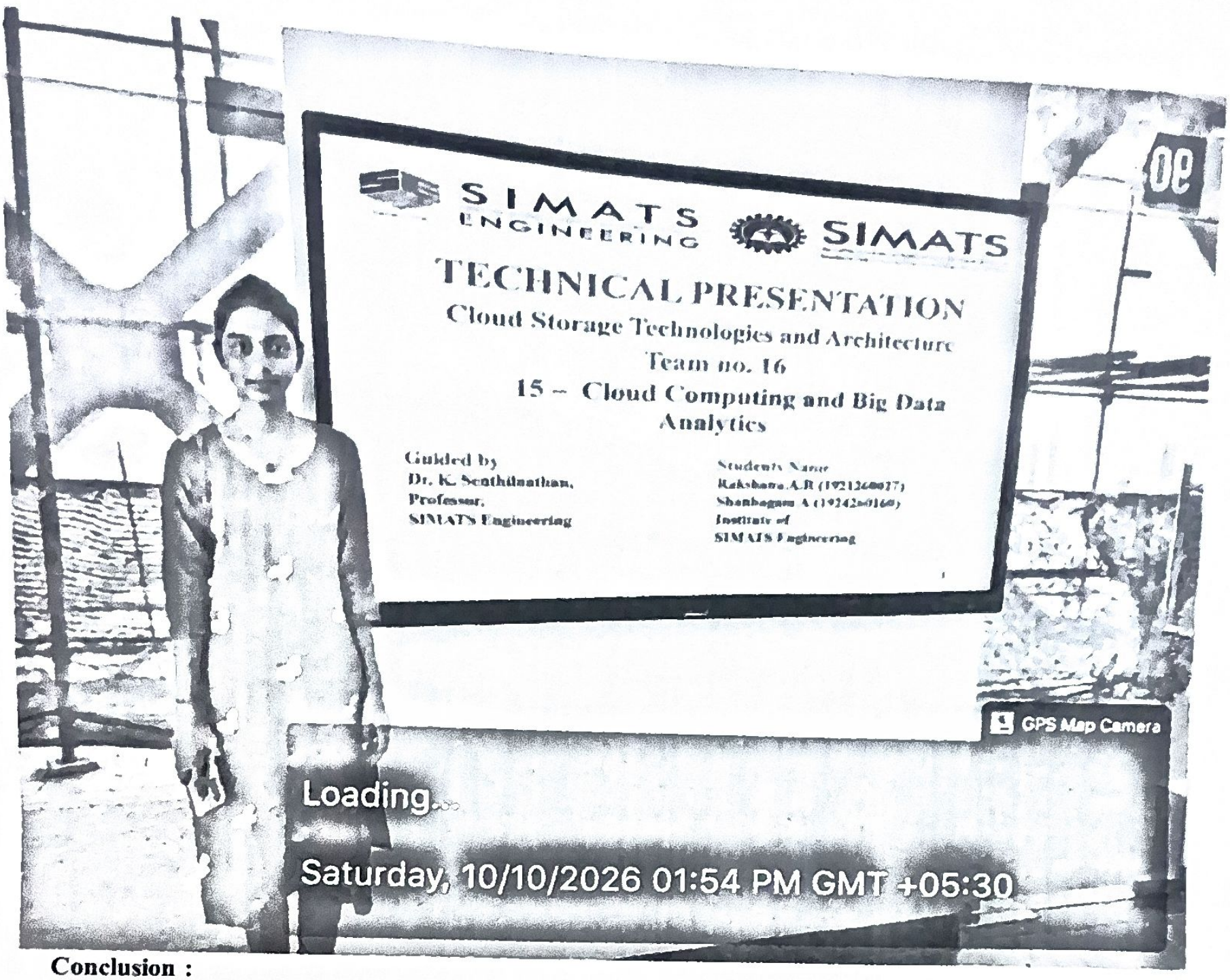
- Launched in 2012, file storage and sync
- 15 GB free, shared across Google account
- Real-time sharing and co-editing
- Version history, offline access, multi-device sync
- Runs on Google's replicated data centers

Conclusion and key takeaway:

Cloud storage enables flexible, network-based data storage. Understanding object, block and file storage, along with architecture, replication and access control, helps explain how cloud services manage data. Amazon S3 demonstrates a practical application of cloud object storage.

Key takeaway: Choose the storage model according to the application's access pattern and data requirements, while applying appropriate security controls.

Photo of the presentation :



Conclusion :

Cloud storage enables flexible, network-based data storage. Understanding object, block and file storage, along with architecture, replication and access control, helps explain how cloud services manage data. Amazon S3 demonstrates a practical application of cloud object storage.